		momentum.
9	**A**	From conservation of momentum, $m_1u_1+m_2u_2 = m_1v_1+m_2v_2$ $2m(5.0)+m(-4.0) = 2mv_1+mv_2$

		Relative speed of approach = relative speed of separation $u_1 - u_2 = v_2 - v_1$ $(5.0) - (-4.0) = v_2 - v_1$ $v_1 = -1.0$ $v_2 = 8.0$
10	**D**	Once gas is released, it expands and achieves atmospheric pressure eventually